RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2024-2025 (Even Sem)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

Date	29th April 2025	Maximum Marks	(10+50) Marks
Course Code	CD363IA	Duration	(30+90) = 120 Min
Sem	VI Sem	CIE-1	
UG/PG	UG	Faculty: AS	

Data Analytics & Visualization(CD363IA)

Scheme & Solution

Q. No.	QUIZ	M
1.	**Describe Data Science as convergence of various knowledge domains.** Data Science is an interdisciplinary field of scientific methods, processes, and systems to extract knowledge or insights from data in various forms structured/unstructured. It is emerging as a convergence of various knowledge domains for effective utilisations of various analysis methods for better output of experts in their activities (indicated Table below)	2

Convergence of various knowledge domains	
Math and Theory	• Statistics, Linear Algebra, Optimization, Time Series, etc.
Applied Algorithms	• Machine Learning, Data Structures, Parallel Algorithms, etc.
Engineering and Technologies	• Storage and computing platforms, statistical tools, etc.
Domain Expertise	• Text, Finance, Images, Economies etc.
Art	• Visualization, Infographics
Best practices and hacks	• Handle missed values in data, transform and represent data, etc.

Data science as convergence of various knowledge domains

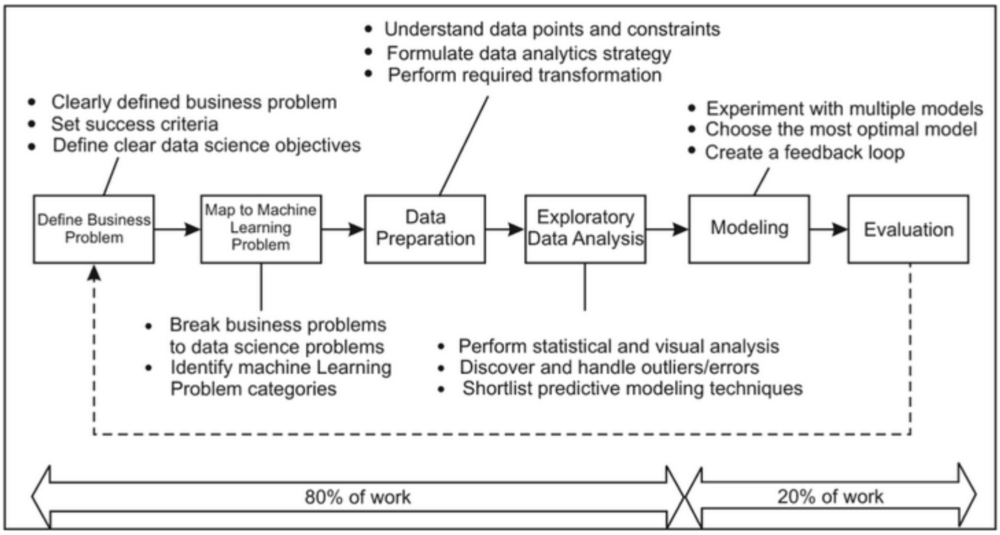
Q. No.	QUIZ	M
2.	**What is Data Wrangling? Give an example.** The process of conversion of data, often using scripting languages, to make it easier to work with is known as Data Wrangling or Data Munging. Eg: If we have 900,000 birthYear values of the format yyyy-mm-dd and 100,000 of the format mm/dd/yyyy and we can write a Perl script to convert the latter to look like the former so that we can use them all together.	2
3.	**Identify the type of Quantitative or Qualitative Data types for the following.:** a) Temperature in Celsius --- 10^0 ---------- Quantitative - Interval b) Age in years ----10 years old ------ Quantitative - Ratio c) Hardness of water --- {soft, moderately hard, hard} -- Qualitative Ordinal d) Eye color --- black, brown ------ Qualitative Nominal	2
4.	**Differentiate between data errors and artifacts during the data cleaning process.** If we view data items as measurements about some aspect of the world, **data errors** represent information that is fundamentally lost in acquisition. The Gaussian noise blurring the resolution of our sensors represents error, precision which has been permanently lost. The two hours of missing logs because the server crashed represents data error: it is information which cannot be reconstructed again. **Artifacts** are generally systematic problems arising from processing done to the raw information it was constructed from. Processing artifacts can be corrected, so long as the original raw data set remains available. These artifacts must be detected before they can be corrected	2

	Data Errors are inaccuracies or inconsistencies in the data caused by issues like human mistakes, faulty sensors, or system glitches (e.g., misspelled names, incorrect values). **Artifacts** are patterns or distortions in data that do not represent the true underlying phenomenon, often introduced by the method of data collection or processing (e.g., repeated noise from a faulty sensor).	
5.	**Spidering** is the process of downloading the right set of pages for analysis. **Scraping** is the fine art of stripping this content from each page to prepare it for computational analysis.	1
6.	An **outlier** is a data point that significantly deviates from the majority of other data points in a dataset. **Outlier detection** is the process of identifying these unusual data points.	1

Q. No.	TEST	M

1.	Discuss about the different types of data - Unstructured, Semi-structured, Meta, and Structured data, bringing out the key features with suitable real-world examples.
	
	Discuss about the different types of data bringing out the key features with suitable real-world examples. --- 2.5 X 4 =10M
	
	1. Unstructured data : Word, PDF, Text, Media Logs.
	2. Semi Structured data : XML data.
	3. Meta Data :Data about data
	4. Structured data : Relational data.
	
	Types of Data in Data Science
	Data can be classified into four broad types based on its structure and format: **Structured**, **Semi-structured**, **Unstructured**, and **Metadata**. Understanding these is essential for choosing appropriate storage, processing, and analysis methods.
	1. Structured Data
	• **Definition**: Data organized in rows and columns, typically stored in relational databases.
	• **Key Features**:
	    o Highly organized and easily searchable using SQL.
	    o Predefined schema.
	• **Examples**:
	    o Customer information stored in a CRM (Name, Age, Email).
	    o Financial transactions in bank databases.
	• **Tools**: SQL, RDBMS, Excel
	
	2. Semi-structured Data
	• **Definition**: Data that does not follow a strict tabular format but contains tags or markers to separate elements.
	• **Key Features**:
	    o Flexible format with partial structure.
	    o Commonly stored in JSON, XML formats.
	• **Examples**:
	    o JSON files from web APIs (e.g., Twitter or YouTube data).
	    o XML documents used in data interchange between systems.
	• **Tools**: NoSQL databases (MongoDB), XML parsers, Python's json library
		10

	3. Unstructured Data
	• **Definition**: Data without a predefined model or format.
	• **Key Features**:
	  ○ Most abundant type of data.
	  ○ Difficult to search, process, and analyze directly.
	  ○ Requires specialized tools and preprocessing.
	• **Examples**:
	  ○ Text documents, emails, audio files, images, videos, social media posts.
	  ○ CCTV footage or customer reviews on e-commerce websites.
	• **Tools**: NLP tools (NLTK, spaCy), multimedia processing libraries, Hadoop, Spark
	4. Metadata
	• **Definition**: Data about data; describes the characteristics of other data.
	• **Key Features**:
	  ○ Helps in identifying, organizing, and managing data.
	  ○ Usually embedded or stored separately.
	• **Examples**:
	  ○ EXIF data in images (camera type, resolution, date).
	  ○ File properties (size, author, date modified).
	  ○ Database schema descriptions (table names, field types).
	• **Tools**: Metadata management tools, data catalogs	

2.	With the aid of neat diagram, discuss the steps of the Data Science Process/ Workflow and illustrate the complete workflow to solve a real-world problem. A neat diagram --- 3M Discuss the steps of the Data Science Process/ Workflow and illustrate the complete workflow to solve a real-world problem. --- 7M  Fig: Seven Steps of Data Science Process	10

Step 1: Frame or define the (business) problem

Step 2: Collect the raw data needed for your problem (and map it to machine learning in case of Big data

Step 3: Data Preparation for process the data for analysis

Step 4: Explore the data (Exploratory Data Analysis (EDA)

Step 5: Perform in-depth analysis (Modelling) and producing prescriptive Business Insights

Step 6: Evaluation

Step 7: Visualisation and Communication of Results of the Analyse

Data Science Process / Workflow

The **Data Science Workflow** refers to a structured sequence of steps followed to extract insights from data and build data-driven solutions. The major stages are as follows:

Steps in the Data Science Workflow

1. Problem Definition

- Understand the business or research problem clearly.
- Define goals and success metrics.
- *Example*: A company wants to reduce customer churn.

2. Data Collection

- Gather data from various sources such as databases, web scraping, IoT sensors, surveys, or logs.
- *Example*: Customer demographics, transaction history, support logs.

3. Data Cleaning & Preprocessing

- Handle missing values, remove duplicates, detect and treat outliers.
- Convert data into suitable formats for analysis.
- *Example*: Removing incomplete customer records or normalizing income values.

4. Data Exploration (EDA)

- Use descriptive statistics and visualization to understand patterns.
- Identify correlations and trends.
- *Example*: Analyzing if churn is related to customer complaints or usage.

5. Feature Engineering

- Create new features or transform existing ones to improve model performance.
- *Example*: Generating features like "average monthly spending" or "complaints per month".

6. Model Building

- Apply machine learning or statistical models such as decision trees, regression, clustering, etc.
- Select algorithms based on data type and objective.
- *Example*: Using logistic regression to predict if a customer will churn.

7. Model Evaluation

- Assess model accuracy using metrics like precision, recall, F1-score, RMSE.
- Perform cross-validation.
- *Example*: The churn model gives 85% accuracy and good recall.

8. Deployment

- Integrate the model into production (web apps, mobile, dashboards).
- Monitor and maintain model performance.
- *Example*: Deploying the churn predictor in the CRM system.

9. Communication of Results

- Share insights and model outcomes using dashboards, reports, or presentations.
- Use visualizations to explain findings to non-technical stakeholders.

Real-World Example: Reducing Customer Churn in a Telecom Company

1. **Problem**: High churn rate affecting revenue.
2. **Data Collection**: Collect data from CRM and call center logs.
3. **Preprocessing**: Remove null entries, encode categorical variables.
4. **EDA**: Discover that churn increases with call complaints.
5. **Feature Engineering**: Create a variable for "customer complaint frequency."
6. **Modeling**: Use logistic regression and decision trees.
7. **Evaluation**: Choose the model with highest AUC score.
8. **Deployment**: Alert system for high-risk customers is built into CRM.
9. **Communication**: Reports are shared with the customer care team for targeted retention.

3a. Compare the features and use cases of Python and R as Data Science toolkits. **05**

Comparison of Python and R as Data Science Toolkits

Feature	Python	R
Ease of Learning	General-purpose language; easy syntax; good for beginners	Designed for statisticians; syntax tailored for statistical computing
Libraries & Packages	Extensive libraries like NumPy, Pandas, Scikit-learn, TensorFlow	Strong packages for statistics and visualization like ggplot2, dplyr
Data Handling	Excellent support for structured and unstructured data	Ideal for statistical analysis, surveys, and academic research
Visualization	Matplotlib, Seaborn, Plotly offer customizable graphics	Superior visualization tools for detailed statistical plots
Integration & Deployment	Integrates well with web apps, APIs, big data tools, and production systems	Less suited for production systems; better for report-based analysis

Use Cases

- **Python** is preferred in:
  - Machine learning and deep learning applications
  - Web-based analytics dashboards
  - Large-scale enterprise solutions
- **R** is preferred in:
  - Academic research and statistical modeling
  - Data visualization-heavy projects
  - Bioinformatics and social science studies

3b. Bring out the importance of Data Visualization and list the conventional Data Visualization methods often used. **05**

Importance of Data Visualization **------- 3M**

Data Visualization plays a critical role in Data Science as it helps to:

- **Simplify complex data**: Converts raw data into visual formats like graphs and charts, making it easier to understand.
- **Identify patterns and trends**: Helps detect relationships, anomalies, and insights that are not obvious in tabular data.
- **Enhance communication**: Enables better storytelling and communication of results to stakeholders with varying technical backgrounds.
- **Support decision-making**: Facilitates informed and faster decisions through intuitive data presentation.

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- **Aid in data exploration**: Allows analysts to explore datasets interactively, which is essential in the initial stages of analysis.

Conventional Data Visualization Methods **---- 2M**

1. **Bar Charts** – Used to compare quantities across different categories.
2. **Line Charts** – Effective for showing trends over time or continuous data.
3. **Pie Charts** – Used to show proportional data or percentage contributions.
4. **Histograms** – Represent frequency distributions of continuous variables.
5. **Scatter Plots** – Used to examine relationships or correlations between two variables.
6. **Box Plots** – Help in identifying data spread, central tendency, and outliers.

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Differentiate between hunting, scraping, and logging as methods of data collection, bringing out the characteristic features, their approach and applications.

Differentiate between hunting, scraping, and logging as methods of data collection, bringing out the characteristic features, their approach and applications. ------ 3.5+3.5+3M

Data Collection Methods in Data Science: Hunting, Scraping, and Logging
Data collection is the foundation of any data science task. Skiena classifies data collection approaches into three broad methods—**hunting**, **scraping**, and **logging**—each with distinct characteristics, approaches, and use cases.

1. Hunting

- **Definition**: Hunting refers to manually searching for data from known, available sources like open datasets, government portals, or published databases.
- **Characteristic Features**:
  - Involves human effort in identifying and downloading existing data.
  - Often involves structured or semi-structured datasets.
- **Approach**:
  - Search for datasets on public repositories (e.g., Kaggle, UCI ML Repository, government websites).
  - Download and integrate them manually.
- **Applications**:
  - Academic research using public health data.
  - Business intelligence from financial records.
  - Prototype models using benchmark datasets.

2. Scraping

- **Definition**: Scraping is the automated extraction of data from websites or HTML pages using scripts or tools.
- **Characteristic Features**:
  - Used when structured APIs are unavailable.
  - Requires knowledge of HTML, DOM structure, and tools like BeautifulSoup or Scrapy.
  - May involve legal and ethical considerations.
- **Approach**:
  - Write scripts to extract relevant data from web pages.
  - Parse and clean the extracted content (e.g., tables, tags).
- **Applications**:
  - Gathering prices from e-commerce websites for competitive analysis.
  - Monitoring news articles or reviews.
  - Collecting job postings from recruitment portals.

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3. Logging

- **Definition**: Logging refers to collecting data generated by systems or users through digital activity tracking (logs).
- **Characteristic Features**:
  - Automatically generated, real-time data.
  - Often unstructured or semi-structured (e.g., server logs, clickstream data).
- **Approach**:
  - Integrate logging frameworks (e.g., Apache Kafka, Logstash).
  - Store logs in scalable systems (e.g., Hadoop, cloud storage).
- **Applications**:
  - User behavior tracking on websites or apps.
  - Network traffic monitoring.
  - Error tracking and system performance analysis.

Comparison Table

Feature	Hunting	Scraping	Logging
Nature	Manual	Automated	Automated (system-driven)
Data Type	Structured/Semi-structured	Unstructured/HTML content	Semi-structured/Unstructured
Effort Level	High (manual search)	Medium (requires coding)	Low (automated once setup)
Tools Used	Browser, Excel	BeautifulSoup, Scrapy, Selenium	Logstash, Fluentd, Kafka
Examples	WHO datasets, Census data	Scraping Flipkart reviews	Server logs, App clickstream logs

5a. Explain the importance of data compatibility when combining datasets from multiple sources. Illustrate with suitable examples how incompatibility issues can be addressed by the following classes of conversions: 05

i) Unit conversion ii) Time/Date Unification.

Importance of Data Compatibility in Merging Datasets **------ 1M**

Data compatibility is **critical when combining datasets from multiple sources**, as inconsistent formats, units, and data types can lead to inaccurate analysis, flawed models, or misinterpretation of results. Skiena emphasizes the need to align datasets so they can be meaningfully integrated.
When datasets differ in measurement units, time formats, or data types, **compatibility issues arise**, making it difficult to perform calculations, comparisons, or visualizations across datasets.

Common Incompatibility Issues and Conversions

i) Unit Conversion **------- 2M**

- **Problem**: Different datasets may use different units for the same attribute (e.g., temperature in Celsius vs. Fahrenheit).
- **Example**:
  - Dataset A: Temperature in **Celsius**
  - Dataset B: Temperature in **Fahrenheit**
- **Solution**: Convert Fahrenheit to Celsius using the formula:
  C=59(F−32)C = \frac{5}{9}(F - 32)
- **Outcome**: Standardizing units enables uniform analysis, such as calculating average temperatures.
-

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ii) Time/Date Unification ------- 2M

Problem: Time may be recorded in various formats or time zones (e.g., DD/MM/YYYY vs. MM-DD-YYYY; IST vs. UTC).
Example:

Dataset A: 2023-12-01 14:30:00 UTC
Dataset B: 01/12/2023 20:00 IST

Solution: Convert all timestamps to a common format and time zone using libraries like Python's datetime, pytz.
Outcome: Prevents errors in time-based analysis, such as tracking events or calculating durations.

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If given a dataset with missing values, describe different strategies for handling these values with suitable examples and evaluate the pros and cons of each.

Handling Missing Values in a Dataset

Heuristic-based imputation: Given sufficient knowledge of the underlying domain, we should be able to make a reasonable guess for the value of certain fields.
Mean value imputation: Using the mean value of a variable as a proxy for missing values is generally sensible. First, adding more values with the mean leaves the mean unchanged, so we do not bias our statistics by such imputation. Second, fields with mean values add a vanilla flavor to most models, so they have a muted impact on any forecast made using the data.
But the mean might not be appropriate if there is a systematic reason for missing data.
Random value imputation: Another approach is to select a random value from the column to replace the missing value. This would seem to set us up for potentially lousy guesses, but that is actually the point. Repeatedly selecting random values permits statistical evaluation of the impact of imputation. If we run the model ten times with ten different imputed values and get widely varying results, then we probably should not have much confidence in the model. This accuracy check is particularly valuable when there is a substantial fraction of values missing from the data set.
Imputation by nearest neighbor: What if we identify the complete record which matches most closely on all fields present, and use this nearest neighbor to infer the values of what is missing? Such predictions should be more accurate than the mean, when there are systematic reasons to explain variance among records.
This approach requires a distance function to identify the most similar records. Nearest neighbor methods are an important technique in data science.
Imputation by interpolation: More generally, we can use a method like linear regression to predict the values of the target column, given the other fields in the record. Such models can be trained over full records and then applied to those with missing values.
Using linear regression to predict missing values works best when there is only one field missing per record. The potential danger here is creating significant outliers through lousy predictions. Regression models can easily turn an incomplete record into an outlier, by filling the missing fields in with unusually high or low values. This would lead downstream analysis
to focus more attention on the records with missing values, exactly the opposite of what we want to do.
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